

MISSOURI S&T
Department of Mathematics & Statistics

Stat 3113 - Sections 101 & 102
Exam 2

SPRING 2025

In-Class Review Session

Key!

1. (8 points) For the multiple choice questions, shade the correct answer.

(a) (2 points) What does "No Memory" mean in the exponential probability distribution?

- ☐ The probability of an event occurring in the future depends on past waiting times.
- ☐ The distribution's shape doesn't change over time as events occur.
- ☒ The probability of an event occurring after a certain time remains constant, regardless of how much time has already passed.
- ☐ None of the above.

(b) (2 points) In a normal distribution, approximately what percentage of data falls within three standard deviations of the mean?

- ☐ 75%
- ☐ 99.5%
- ☐ 68%
- ☒ 99.7%

(c) (2 points) The significance level of a test α controls the probability that you:

- ☒ Reject the null, when it's true.
- ☐ Fail to reject the null, when it's false.
- ☐ Reject the null, when it's false.
- ☐ Fail to reject the null, when it's true.

(d) (2 points) Which of the following decreases the width of a confidence interval for the population mean? **Select all that apply.**

- ☒ Increasing the sample size
- ☐ Decreasing the sample size
- ☒ Increasing the significance level
- ☐ Decreasing the significance level
- ☐ Increasing the confidence level
- ☒ Decreasing the confidence level
- ☐ Increasing the mean
- ☐ Decreasing the mean

2. (10 points) Indicate whether the following statements are true or false.

- (a) ☐ True ☒ False The exponential distribution can be used to model symmetric data around the mean.
- (b) ☒ True ☐ False As the degrees of freedom increase, the Student's t -distribution approaches the standard normal distribution.
- (c) ☐ True ☒ False The Central Limit Theorem ensures that the population distribution is always normal.
- (d) ☒ True ☐ False The P value is the probability of obtaining a value for the test statistic that is extreme or more extreme than the value actually observed.
- (e) ☒ True ☐ False The probability to correctly reject a false null hypothesis is called the Power of the test

3. (16 points) In the field of materials engineering, a specific parameter, denoted as X , represents the tensile stress experienced by a material during a testing process. The behavior of X follows a probability density function (pdf) characterized by the following expression:

$$f(x) = \begin{cases} cx^2, & |x| \leq 1 \quad (\Rightarrow -1 \leq X \leq 1) \\ 0, & \text{otherwise} \end{cases}$$

- (a) (4 points) To ensure accurate stress analysis, engineers need to determine the constant c that characterizes the pdf for material stress. This constant is crucial in understanding the behavior of the material under test conditions.

Answer = $\int_{-1}^1 cx^2 dx = c \left[\frac{1}{3} x^3 \right]_{-1}^1 = \frac{2}{3} c = 1 \Rightarrow c = \frac{3}{2}$

- (b) (4 points) Engineers are interested in calculating the expected value (mean) of the material stress, represented as $E(X)$. This allows them to predict the typical or average stress the material experiences during testing.

Why? $E[X] = \int_{-1}^1 x f(x) dx = \int_{-1}^1 \frac{3}{2} x^3 dx = \frac{3}{2} \int_{-1}^1 x^3 dx = 0$

$\hookrightarrow g(x) = x^3$ is odd on a symmetric interval

Answer = 0

- (c) (4 points) Understanding the variability in stress levels is essential. Engineers need to determine the variance, $V(X)$, to quantify the spread or dispersion of stress values.

$$V(X) = E[X^2] - E[X]^2 = E[X^2] \text{ since } E[X] = 0$$

$$= \int_{-1}^1 \frac{3}{2} x^4 dx = \frac{3}{2} \left[\frac{1}{5} x^5 \right]_{-1}^1 = \frac{3}{5}$$

Answer =

$$3/5$$

- (d) (4 points) When designing materials for specific applications, it's crucial to assess extreme scenarios. Engineers aim to find the probability that material stress (X) exceeds or equals a critical value of $\frac{3}{4}$, denoted as $P(X \geq \frac{3}{4})$. This information guides material selection for applications with stringent stress requirements.

$$P(X \geq \frac{3}{4}) = \int_{\frac{3}{4}}^1 f(x) dx = \frac{3}{2} \int_{\frac{3}{4}}^1 x^2 dx = \frac{3}{2} \left[\frac{1}{3} x^3 \right]_{\frac{3}{4}}^1$$

Answer =

$$37/128$$

- (e) (4 points) Determine the median (Q_2) of the material stress distribution represented by the pdf.

$$\text{Solve } F(x_m) = 0.5$$

$$\int_{-1}^{x_m} f(x) dx = \frac{3}{2} \left[\frac{1}{3} x^3 \right]_{-1}^{x_m} = \frac{1}{2} (x_m^3 + 1) = \frac{1}{2} \Rightarrow x_m^3 + 1 = 1$$

Answer = $x_m = 0.$

- (f) (4 points) Calculate the first quartile (Q_1) of the material stress distribution.

$$\text{Solve } F(x_{25}) = 0.25$$

$$\int_{-1}^{x_{25}} f(x) dx = \frac{3}{2} \left[\frac{1}{3} x^3 \right]_{-1}^{x_{25}} = x_{25}^3 + 1 = \frac{1}{2}$$

Answer = $x_{25} \approx -0.63$

$$\Rightarrow x_{25} = \sqrt[3]{-\frac{1}{2}}$$

4. (14 points) Let X denote the waiting time in minutes until the first customer arrives in a certain shop and assume that $X \sim \text{Exp}(\lambda = 1/3)$.

(a) (4 points) What is the probability that the shopkeeper will have to wait more than 5 minutes for the arrival of the first customer?

$$P(X > 5) = e^{-\frac{1}{3} \times 5}$$

Answer =

0.1889

(b) (5 points) What is the median time until the first arrival?

$$\text{Solve } F(x_m) = 0.5$$

$$\Rightarrow 1 - e^{-\frac{1}{3}x_m} = 0.5$$

$$\Rightarrow x_m = -3 \ln(0.5)$$

Answer =

2.0794

(c) (5 points) Given that the shopkeeper has waited for 5 minutes without any customer arrivals, what is the probability that they will have to wait for an additional 3 minutes for the first customer to arrive?

$$\begin{aligned} P(X > 5 + 3 \mid X > 5) &= P(X > 3) \\ &= e^{-\frac{1}{3}(3)} \end{aligned}$$

Answer =

0.3679

5. (10 points) In a food processing plant, the weight of sugar in a standard package is known to be normally distributed with a mean weight (μ) of 15 grams and a standard deviation (σ) of 4 grams.

For quality control, 25 packages of sugar are randomly selected for testing.

- (a) (2 points) Does the Central Limit Theorem apply here? Justify your answer in just one sentence.

No, $n < 30$.

- (b) (3 points) What is the standard error of the sample mean?

$$SE(\bar{X}) = \frac{\sigma}{\sqrt{n}} = \frac{4}{\sqrt{25}} = \frac{4}{5}$$

Answer =

0.8

- (c) (5 points) Calculate the probability that the sample mean weight is less than 16 grams.

$$P(\bar{X} \leq 16) = P\left(\frac{\bar{X} - 15}{0.8} \leq \frac{16 - 15}{0.8}\right) = P(Z \leq 1.25)$$

From CDF Table,

Answer =

0.8944.

6. (30 points) Many grocery stores and large retailers, including Walmart, have installed self-checkout systems so shoppers can scan their own items and cash out themselves. Evidence suggests these machines are cost-effective if more than 100 customers on average use the self-checkout per day. The sample consists of the number of customers using the service over a period of 15 days at the Walmart on Highway 544 in Surfside, South Carolina. This data was collected by Caleb, Oliver and Breanna, who were tasked with evaluating the efficiency of the self-checkout system at this specific location.

Is it reasonable to conclude that the machine is cost effective? Test at level 0.05

Below is an output from JMP. Use the output below to answer the questions that follow.

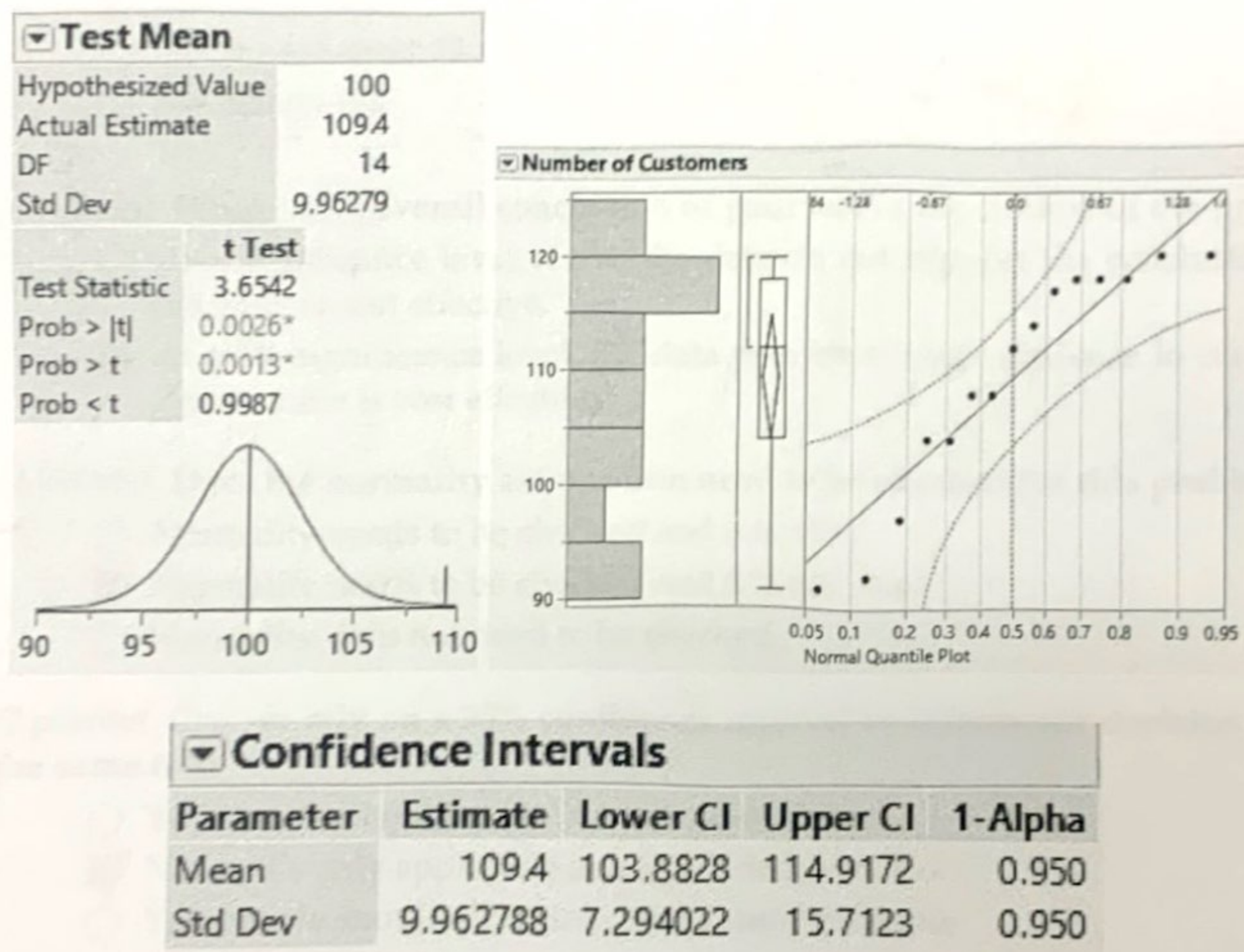


Figure 1: JMP Output.

- (a) (4 points) Which specific statistical test was performed?

- ☒ T-test for single mean (σ unknown)
- ☐ Z-test for single mean
- ☐ Paired-data T-test
- ☐ Two-Sample Independent T-test

- (b) (4 points) Select the appropriate hypothesis.

☐ $H_0 : \mu = 100$
 $H_1 : \mu < 100$

☐ $H_0 : \mu = 100$
 $H_1 : \mu \neq 100$

☒ $H_0 : \mu = 100$
 $H_1 : \mu > 100$

(c) (4 points) Obtain the test statistic from the output provided.

Test Statistic =

3.1542

(d) (4 points) Identify the appropriate p-value from the output provided.

P value =

0.0013

(e) (4 points) Based on the p-value in part(d), should H_0 be rejected?

- ☒ Yes, reject H_0 .
☐ No, do not reject H_0 .
☐ No, accept H_0 .

$\rightarrow (P \text{ value} < \alpha)$

(f) (4 points) What is the overall conclusion of your test in the context of the problem?

- ☐ At a significance level of 5%, the data do not support the conclusion that the machine is cost effective.
☒ At a 5% significance level, the data provide enough evidence to conclude that the machine is cost effective.

(g) (4 points) Does the normality assumption need to be checked for this problem?

- ☐ Normality needs to be checked and it is okay.
☒ Normality needs to be checked and it is not okay.
☐ Normality does not need to be checked.

(h) (2 points) Can we rely on a 95% confidence interval to inform our decision regarding the same test?

- ☐ Yes, and we should reach the same conclusion.
☒ No, as it's only applicable in a two-sided scenario.
☐ Yes, but we shouldn't anticipate the same outcome.